



CYBER SECURITY & ETHICAL HACKING INTERNSHIP PROGRAM

PROJECT 01 · CASE FILE CA-ETI-01

Email Threat Investigation & Analysis

Every inbox tells a story — find the lie hiding in the headers.

CASE BRIEFING

Thirteen (13) emails have been recovered from the mailbox of **Mr. Aditya Rao, Senior Finance Executive at Solvex Industries Pvt. Ltd.**, after the finance team flagged unusual activity and a suspected fraud attempt. You are being brought in as the investigating Cyber Security Analyst.

Each of the following 13 pages contains one recovered email exactly as received — subject, sender details, technical headers and full message body, along with any attachment or link found in it. Some are genuine, some are clearly malicious, and a few are deliberately tricky. Think like a real investigator: read every detail, question what looks "off," and base your judgement on evidence, not assumption.

YOUR OBJECTIVES

1. Go through all 13 emails (Pages 2–14) one at a time — do not skim.
2. Study the sender address, Reply-To, Return-Path, and the SPF / DKIM / DMARC / encryption results for each email.
3. Read the full message body and identify any social engineering tactics used (urgency, fear, authority, curiosity, greed, etc).
4. Examine any attachment or link mentioned and note why it is safe or risky.
5. Classify each email as **SAFE**, **SUSPICIOUS**, or **MEDIUM RISK**, with clear, evidence-based reasoning for your decision.
6. For every suspicious or medium-risk email, explain what could happen if the recipient clicked the link / opened the attachment / acted on the request — and what action should be taken instead.
7. Compile your complete findings into one clean, professional PDF investigation report.

YOUR REPORT MUST INCLUDE

Section	What to Cover
Executive Summary	Overall findings in brief — how many emails were safe / suspicious / medium risk
Email-by-Email Analysis	Classification + detailed reasoning for each of the 13 emails
Indicators of Compromise (IOCs)	Suspicious domains, IPs, links and attachments identified
Recommended Actions	What the employee / organisation should do for each risky email

A NOTE BEFORE YOU BEGIN

This project is meant to build your real, practical analytical skills. You may use AI tools to research or clear doubts, but the final analysis and judgement must be your own. A fully AI-written report defeats the purpose of this internship — you are here to build a skill, not just produce a document.

REPORT SUBMISSION

Once your report is complete, submit it as a PDF using the official submission form below.

[\[CLICK HERE TO SUBMIT YOUR REPORT \]](#)



Karndeeep Baror

Cyber Crime Investigator & Threat Intelligence



Connect on LinkedIn — [\[Click Here to Connect \]](#)



WhatsApp for guidance: +91 9509622197

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Password Expiry Reminder – Action Needed Within 5 Days

HEADER & TECHNICAL SNAPSHOT

From	IT Helpdesk – Solvex Industries <ithelpdesk@solvexindustries.com>
Reply-To	ithelpdesk@solvexindustries.com
Return-Path	ithelpdesk@solvexindustries.com
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	—
Date & Time	03 July 2026, 10:12 AM IST
Received From	mail.solvexindustries.com (192.168.14.22)
SPF	Pass
DKIM	Pass
DMARC	Pass
Encryption (TLS)	Yes (TLS 1.3)

EMAIL BODY

Dear Aditya,

This is an automated reminder from the IT Helpdesk. Our records show that your network account password is due to expire in 5 days, as per the company's 90-day password rotation policy.

To avoid any interruption to your email or system access, please update your password before the expiry date. You can do this at any time by logging into the internal IT Service Portal from your office desktop or the VPN-connected laptop and navigating to Account Settings > Change Password.

If you face any difficulty resetting your password, please raise a ticket through the IT Service Portal or contact the Helpdesk extension 204 during working hours. Do not share your password with anyone, including IT staff, over email, call or chat.

Regards,

IT Helpdesk Team

Solvex Industries Pvt. Ltd.

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

URGENT: Your Mailbox Storage is FULL – Verify Now to Avoid Suspension

HEADER & TECHNICAL SNAPSHOT

From	IT Support Desk <support@solvex-industries-helpdesk.com> recovery-team@mail-secure-verify.net
Reply-To	
Return-Path	bounce@mail-secure-verify.net
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	allstaff-list223@solvex-industries-helpdesk.com
Date & Time	05 July 2026, 11:47 PM IST
Received From	unknown-host-193.41.77.108.static-cloud.ru (193.41.77.108)
SPF	Fail
DKIM	Fail
DMARC	Fail (Reject policy overridden)
Encryption (TLS)	No (Unencrypted)

EMAIL BODY

Dear User,

Your mailbox has exceeded its allocated storage limit of 5GB. As per policy, mailboxes exceeding this limit for more than 24 hours will be automatically suspended and all incoming mail will be permanently rejected.

To avoid suspension of your account, you are required to verify your identity IMMEDIATELY by clicking the secure link below and confirming your login credentials. This process takes less than 2 minutes.

Failure to verify within 24 hours will result in permanent loss of access to your mailbox and all stored data. This is a final notice.

Thank you for your prompt action.

IT Support Team

EMBEDDED LINK DETECTED

hxxp://solvex-industries-helpdesk.verify-account-secure.com/mailbox/confirm?user=aditya.rao

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Invoice SI-4471 & Updated Bank Details for Future Payments

HEADER & TECHNICAL SNAPSHOT

From	Ramesh Kulkarni – Accounts, Apex Steel Traders <ramesh.kulkarni@apex-steeltraders.co> accounts.payments@apexsteel-billing.com
Reply-To	
Return-Path	ramesh.kulkarni@apex-steeltraders.co
To	aditya.rao@solvexindustries.com
Cc	priya.menon@solvexindustries.com
Bcc	—
Date & Time	07 July 2026, 3:28 PM IST
Received From	mail.apex-steeltraders.co (103.22.8.190)
SPF	Pass (for apex-steeltraders.co)
DKIM	Pass
DMARC	Pass
Encryption (TLS)	Yes (TLS 1.2)

EMAIL BODY

Dear Aditya,

Please find attached Invoice SI-4471 for the steel coil consignment supplied to your Nagpur unit last month, amounting to Rs. 4,85,600/-.

As part of our internal banking audit, our current account has been temporarily restricted. Kindly note that all payments from this point forward should be made to our NEW account, details of which are mentioned in the attached invoice, and NOT our previously shared account. This change is effective immediately.

We would appreciate it if this payment could be processed on priority, as our finance team needs to close this quarter's books by the 10th. Please confirm once the transfer has been initiated so we can update our records.

Regards,

Ramesh Kulkarni

Accounts Department, Apex Steel Traders

FILE

Invoice_SI-4471_Apex_Steel.pdf
PDFDocument • 812KB

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Revised Leave & Work-From-Home Policy – Effective 15 July 2026

HEADER & TECHNICAL SNAPSHOT

From	Human Resources – Solvex Industries <hr@solvexindustries.com>
Reply-To	hr@solvexindustries.com
Return-Path	hr@solvexindustries.com
To	all-employees@solvexindustries.com
Cc	—
Bcc	—
Date & Time	08 July 2026, 9:05 AM IST
Received From	mail.solvexindustries.com (192.168.14.19)
SPF	Pass
DKIM	Pass
DMARC	Pass
Encryption (TLS)	Yes (TLS 1.3)

EMAIL BODY

Dear Team,

As communicated in last month's town hall, the revised Leave and Work-From-Home Policy will come into effect from 15th July 2026. Key changes include an increase in casual leave from 8 to 10 days per year, and a structured hybrid work option of up to 2 WFH days per week subject to manager approval.

The complete policy document has been uploaded to the HR Portal under Policies > Leave Policy 2026. Please log in using your employee credentials to review the full document.

If you have questions regarding the revised policy, please reach out to your respective HR Business Partner or attend the Q&A session scheduled for 12th July at 4:00 PM in Conference Room 2.

Warm regards,

Human Resources Team

Solvex Industries Pvt. Ltd.

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Confidential – Immediate Wire Transfer Required (Time Sensitive)

HEADER & TECHNICAL SNAPSHOT

From	Rajeev Malhotra (Managing Director) <rajeev.malhotra.md@gmail-corpmail.com>
Reply-To	rajeev.malhotra.md@gmail-corpmail.com
Return-Path	rajeev.malhotra.md@gmail-corpmail.com
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	—
Date & Time	09 July 2026, 1:14 PM IST
Received From	smtp-relay-88.freemailhost.io (45.137.22.9, Netherlands)
SPF	None (domain has no SPF record)
DKIM	None
DMARC	None
Encryption (TLS)	No

EMAIL BODY

Aditya, I'm currently in a board meeting with our overseas investors and cannot take calls right now. I need you to process an urgent international wire transfer of USD 18,500 to close a confidential acquisition deal today. This is extremely time sensitive and must be kept strictly between us until I announce it to the board. I will send the beneficiary bank details in my next email. Please confirm you can initiate this within the next hour and let me know once done. Do not discuss this with Priya or anyone in Accounts for now — I want to keep this quiet until the deal is signed. I'm counting on you for this one. Talk soon. Rajeev

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

'Q1_Vendor_Compliance_Report.xlsx' has been shared with you

HEADER & TECHNICAL SNAPSHOT

From	Google Drive <drive-shares-noreply@google.com>
Reply-To	drive-shares-noreply@google.com
Return-Path	drive-shares-noreply@google.com
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	—
Date & Time	09 July 2026, 6:40 PM IST
Received From	mail-sor-f89.google.com (209.85.220.89)
SPF	Pass
DKIM	Pass
DMARC	Pass
Encryption (TLS)	Yes (TLS 1.3)

EMAIL BODY

priya.menon@solvexindustries.com shared a file with you

Q1_Vendor_Compliance_Report.xlsx

Priya added a note: "Hi Aditya, sharing the vendor compliance sheet we discussed in today's finance sync. Can you review the highlighted rows before Friday's audit call? Thanks!"

Open in Google Sheets

You are receiving this email because someone shared a file or folder located in Google Drive with you. Google Drive is a place to keep and share all your stuff, photos, stories, designs, drawings, recordings, videos — whatever.

EMBEDDED LINK DETECTED

<https://docs.google.com/spreadsheets/d/1kQ3z-example-fileid-9f2/edit?usp=sharing>

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Unusual sign-in activity detected on your Microsoft account

HEADER & TECHNICAL SNAPSHOT

From	Microsoft account team <security-noreply@microsoft-online.com>
Reply-To	no-reply@microsoft-online.com
Return-Path	bounce@microsoft-online.com
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	—
Date & Time	10 July 2026, 2:03 AM IST
Received From	host-77-91-134-6.vpn-exit.net (77.91.134.6, unknown origin)
SPF	Fail
DKIM	Fail
DMARC	Fail
Encryption (TLS)	No

EMAIL BODY

Microsoft account

We detected an unusual sign-in attempt on your account from a new device in Lagos, Nigeria on 10 July 2026 at 1:58 AM. If this wasn't you, your account may be compromised.

For your security, we have temporarily limited access to your account. To restore full access, please verify your identity within 12 hours by confirming your account details using the secure link below.

If you don't recognise this activity, we strongly recommend you review your account immediately to prevent unauthorized access to your email, OneDrive files and Teams messages.

Thanks,

The Microsoft account team

EMBEDDED LINK DETECTED

hxxp://login.microsoft-online.com/secure/verify-identity?ref=aditya.rao

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Exciting Career Opportunity – Senior Finance Role (Immediate Response Needed)

HEADER & TECHNICAL SNAPSHOT

From	Neha Kapoor, HR Consultant <neha.kapoor.hr@talentbridge-recruiters.net>
Reply-To	neha.kapoor.hr@talentbridge-recruiters.net
Return-Path	neha.kapoor.hr@talentbridge-recruiters.net
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	—
Date & Time	10 July 2026, 11:20 AM IST
Received From	mail.talentbridge-recruiters.net (156.213.44.71, Singapore)
SPF	Pass (for talentbridge-recruiters.net)
DKIM	Pass
DMARC	None (no policy published)
Encryption (TLS)	Yes (TLS 1.2)

EMAIL BODY

Dear Candidate,

Your profile has been shortlisted for a Senior Finance Manager position with a leading MNC client of ours, offering a salary 40% higher than industry average. We are impressed by your experience and would like to move forward quickly as the client wants to close this position this week.

Attached is the detailed Job Description along with the client company profile and compensation structure for your review. Please go through it and reply within 24 hours confirming your interest, along with a scanned copy of your latest salary slip and PAN card for our internal verification process.

Looking forward to your quick response so we can schedule your interview at the earliest.

Best regards,

Neha Kapoor

Senior HR Consultant, TalentBridge Recruiters

FILE

JD_SeniorFinanceManager_Client.zip

ZIPArchive(compressed) • 2.4MB

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Your TallyPrime Cloud Subscription Renewal Invoice – INV-88213

HEADER & TECHNICAL SNAPSHOT

From	Tally Solutions Billing <billing@tallysolutions.com> billing@tallysolutions.com
Reply-To	
Return-Path	billing@tallysolutions.com
To	aditya.rao@solvexindustries.com
Cc	accounts@solvexindustries.com
Bcc	—
Date & Time	11 July 2026, 10:00 AM IST
Received From	mta-out.tallysolutions.com (203.0.113.44)
SPF	Pass
DKIM	Pass
DMARC	Pass
Encryption (TLS)	Yes (TLS 1.3)

EMAIL BODY

Dear Customer,

This is to notify you that your TallyPrime Cloud annual subscription (License ID: TP-2291-SOLVEX) is due for renewal on 20 July 2026. The renewal invoice INV-88213 for Rs. 32,500/- (inclusive of GST) is attached for your records.

Payments can be made through your existing registered payment method on the Tally Customer Portal, or via NEFT/RTGS to the bank account already on file with our billing team — no bank details have changed.

If you wish to modify your plan or add additional user licenses before renewal, please contact your Tally partner or our billing support at billing@tallysolutions.com.

Thank you for being a valued customer.

Tally Solutions Pvt. Ltd. – Billing Team

FILE

INV-88213_TallyPrime_Renewal.pdf
PDFDocument • 146KB

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

CONGRATULATIONS!!! YOU HAVE WON USD 1,000,000 IN THE INTERNATIONAL EMAIL LOTTERY

HEADER & TECHNICAL SNAPSHOT

From	INTERNATIONAL LOTTERY BOARD <claims.department009@yandex-mailer.com>
Reply-To	agent.williams.claims@outlook-verify.info
Return-Path	claims.department009@yandex-mailer.com
To	undisclosed-recipients
Cc	—
Bcc	aditya.rao@solvexindustries.com
Date & Time	11 July 2026, 4:52 PM IST
Received From	unknown-relay-node4.freehosting-mail.com (185.220.101.44, Netherlands)
SPF	Fail
DKIM	Fail
DMARC	Fail
Encryption (TLS)	No

EMAIL BODY

ATTENTION WINNER,

We are pleased to inform you that your email address has WON the sum of USD 1,000,000.00 (One Million US Dollars Only) in the International Email Lottery Programme 2026, held under the supervision of the European Lottery Commission.

To claim your prize, you are required to contact our Claims Agent immediately with your full name, address, phone number, and a copy of your ID for verification. A small processing and courier fee of USD 250 will be required to release your winning certificate and cheque.

Please respond within 48 hours failure of which your prize will be forfeited and awarded to another participant.

Congratulations once again!

Dr. James Williams, Claims Coordinator

EMBEDDED LINK DETECTED

hxxp://claim-your-prize-now.lottery-verify.info/claim?ref=44921

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Quick request before my flight

HEADER & TECHNICAL SNAPSHOT

From	Suresh Iyer (CFO) <suresh.iyer@solvexindustries.com> s.iyer.cfo.travel@outlook.com
Reply-To	
Return-Path	suresh.iyer@solvexindustries.com
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	—
Date & Time	12 July 2026, 8:31 AM IST
Received From	mail.solvexindustries.com (192.168.14.19) — header shows internal relay, but Reply-To routes externally
SPF	Pass (envelope sender matches solvexindustries.com)
DKIM	Pass
DMARC	Pass
Encryption (TLS)	Yes (TLS 1.3)

EMAIL BODY

Aditya,

About to board my flight to Mumbai and my corporate card isn't working at the airport lounge. Can you do me a favour and buy 5 Amazon gift cards of Rs. 5,000 each from the online store, then send me the card codes here — I'll reimburse you as soon as I land. Need it in the next 20 minutes before boarding closes.

Please don't call the office landline about this, my phone network is patchy while boarding, email is easier to check. Really appreciate this, will make it up to you.

Suresh

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Reminder: Mandatory Security Awareness Training – Due This Friday

HEADER & TECHNICAL SNAPSHOT

From	Information Security Team – Solvex Industries <infosec@solvexindustries.com> infosec@solvexindustries.com
Reply-To	
Return-Path	infosec@solvexindustries.com
To	all-employees@solvexindustries.com
Cc	—
Bcc	—
Date & Time	12 July 2026, 9:15 AM IST
Received From	mail.solvexindustries.com (192.168.14.19)
SPF	Pass
DKIM	Pass
DMARC	Pass
Encryption (TLS)	Yes (TLS 1.3)

EMAIL BODY

Dear Colleague,

This is a reminder that the mandatory Q3 Security Awareness Training module, covering phishing identification, password hygiene and safe attachment handling, must be completed on the internal Learning Portal by this Friday, 17 July 2026.

As highlighted in recent incidents across the industry, email-based fraud continues to be one of the top risks to organisations. This training will take approximately 25 minutes to complete and includes a short assessment at the end.

To access the module, log in to learn.solvexindustries.com using your employee ID. If you experience any login issues, please raise a request with the IT Helpdesk through the official portal.

Regards,

Information Security Team

Solvex Industries Pvt. Ltd.

RECOVERED MAILBOX EVIDENCE — CASE FILE CA-ETI-01

Pending Invoice – Please Review & Process on Priority

HEADER & TECHNICAL SNAPSHOT

From	Accounts Receivable <accounts.receivable@shreeganesh-logistics.in> accounts.receivable@shreeganesh-logistics.in
Reply-To	
Return-Path	mailer@shreeganesh-logistics.in
To	aditya.rao@solvexindustries.com
Cc	—
Bcc	—
Date & Time	13 July 2026, 5:58 PM IST
Received From	webmail-host-4.sharedhosting-cluster.com (41.203.88.17, unrelated hosting cluster)
SPF	Fail (domain not authorised to send)
DKIM	None
DMARC	Fail
Encryption (TLS)	No

EMAIL BODY

Dear Sir,

Please find attached the pending transport invoice for the last three consignments dispatched to your Pune warehouse. Kindly review and process the payment at the earliest as our accounts are overdue for closing this month.

The attached file also contains the delivery challans and updated GST details for your reference. Please enable editing/macros if prompted so all the invoice tables display correctly, as the file was generated on an older billing system.

Kindly confirm receipt of this email and expected date of payment.

Thanking you,

Accounts Receivable Team

Shree Ganesh Logistics

FILE

Pending_Invoice_Challan_Details.docm
Word Document (Macro-Enabled) • 1.1 MB

EMBEDDED LINK DETECTED

hxxp://shreeganesh-logistics.in.invoice-view-secure.com/download?file=inv2207